

JOYCE SUDI

Ph.D. Candidate | R&D Engineer in Automatic Control & Robotics

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Ph.D.-level engineer with expertise in control systems, robotics, estimation, and autonomous systems. Interested in industrial R&D roles focused on advanced control, navigation, and robotic autonomy.

EXPERIENCE

Autonomous Driving Research Engineer @ Stellantis

Since February 2024 | Poissy, France

- Develop control algorithms for autonomous driving and ADAS with applications to vehicle dynamics and motion control.
- Contribute to model-based decision and control strategies for autonomous vehicles in an industrial R&D environment.

R&D Engineer Intern @ Stellantis

Feb. 2023 - Aug. 2023 | Poissy, France

3D Vehicle Digital Twin for Load Identification Using Real Driving Data

- Developed a load identification method for vehicle suspension systems using an augmented and constrained Extended Kalman Filter (EKF) applied to a nonlinear half-vehicle multibody model.
- Industrialized the methodology in a Python environment to support its integration into the company's engineering workflow.

Robotics Engineer @ Clubelek - INSA Lyon

Sept. 2020 - Feb. 2023 | Villeurbanne, France

- Contributed to the development of a competition robot for the French Robotics Cup within Clubelek, INSA Lyon's mechatronics and robotics club.
- Worked on module coordination and systems integration.
- Contributed to robot localization and path planning in a ROS-based robotics environment using Python and C/C++.

EDUCATION

Ph.D. Candidate in Automatic Control @ IRIMAS UR 7499 - Univ. de Haute-Alsace

2024 - 2027 | Mulhouse, France

Research area : Decision and Control Systems for Autonomous Vehicles; Multi-Objective Optimal Control.

Thesis title: Design of multi-controller architectures for automated driving in heterogeneous and critical scenarios

Selected training:

- Learning-Based Model Predictive Control(Prof. Dr. M. Zeilinger, EECI-IGSC)
- Model Predictive Control and Reinforcement Learning (Prof. Dr. M. Diehl)
- Adaptive control (Prof. Dr. R. Orjuela)

MEng Mechanical Engineering @ INSA Lyon

2020 - 2023 | Villeurbanne, France

Major : Mechatronics and systems

Selected coursework: Vehicle Dynamics, Optimization, Multi-Physical Systems Modeling, Identification and Active Control of Structures, Actuator Architecture, Design and Sizing

LINKS

GitHub: github.com/joycesudi
LinkedIn: linkedin.com/in/joycesudi
Google Scholar: [profile](#)
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SKILLS

Mathematics & Control Engineering

- Nonlinear & Optimal Control
- Data-driven Control
- Optimization & Metaheuristics
- Modeling & Simulation of Dynamical Systems
- State Estimation
- Signal Processing

Programming Languages

- Proficient: C/C++, Java (Android Development), Bash
- Experienced: MATLAB/Octave, Python
- Markup: $\LaTeX$ / $\TeX$, XML, Markdown

Libraries & Frameworks

- Numerical & ML: NumPy, SciPy, Matplotlib, pandas, scikit-learn, TensorFlow, PyTorch
- Robotics: ROS & ROS 2

Tools & Platforms

- OS: GNU Linux (Ubuntu, Debian), MS Windows.
- Version Control: Git (GitHub, GitLab)
- Electronics and embedded systems: Arduino, STM32, Raspberry Pi, KiCAD

Languages

French (native) ●●●●●●●●
English (Linguaskill 180/180) ●●●●●●●●
Swahili (B1) ●●●●●●●●
German (A1/A2 courses) ●●●●●●●●

MSc Mechanical Engineering – Visiting Student (Exchange Program) @ EPFL

2021 - 2022 | Lausanne, Switzerland

Selected coursework: Basics of mobile robotics, Basics of robotics for manipulation, Nonlinear control, Multivariable control, Embedded motor control.

SELECTED PROJECTS

Design and Development of a Quadcopter Testbed (Master's Thesis) @ Ampère Laboratory - CNRS UMR 5005

Sept. 2022 - Feb. 2023 | Villeurbanne, France

- Modeled and simulated quadcopter flight dynamics for educational and experimental use in aerospace systems control.
- Designed and implemented control laws for UAV stabilization.
- Built and validated a functional prototype through testing and experimental verification.

CubeSat Prototyping and Testing @ Mechanical Engineering Department - INSA Lyon

Feb. 2022 - Jun. 2023 | Villeurbanne, France

- Designed the on-board computer and embedded electronics architecture of a university CubeSat.
- Contributed to prototyping and validation activities for the satellite's electronic subsystems.

PUBLICATIONS

Context-Aware Hierarchical MPC for Autonomous Vehicle Path Tracking

Under review at 2026 IEEE International Conference on Systems, Man, and Cybernetics (SMC 2026)

Joyce Sudi Bin-Ali, Imane Mahtout, Michel Basset, Rodolfo Orjuela

Multi-Scenario Multi-Objective Predictive Controller for Lateral Control of Autonomous Vehicles

Published article in the IFAC IAV 2025 conference proceedings – doi.org/10.1016/j.ifacol.2025.07.013

Joyce Sudi Bin-Ali, Imane Mahtout, Michel Basset, Rodolfo Orjuela

PATENTS

Multi-Scenario Control Strategy for Autonomous Vehicles

Patent application pending – awaiting conflict review

ACTIVITIES & INTERESTS

- Volleyball and cycling
- Chess
- Reading, especially classical literature and science fiction